5.6 INITIAL CAL/BIT AND TOA DELAY SETTINGS

INITIAL SETUP:

Tools and Special Tools
920-00636-100, Site Calibration Kit

References
5 System Configuration WP
6.2 Calibration WP

Materials/Parts
None

Equipment Condition
5.5 Input Survey Data WP Completed
Equipment Powered On
Generator or Shore power applied

Personnel Required
Technician (2)

GENERAL INFORMATION

- This procedure is designed to set initial Cal/BIT and frequency settings. Before TOA delay settings can be calculated, survey data must be entered, and system must be passing BIT.
- The following procedure will need to be repeated for the second rack if system is configured with dual racks.
- Switch to the rack being configured on the Remote Control Unit (RCU).
- This subsection deals with positioning the Cal/BIT pulse in the “framing” window. This will allow the system to take measurements from the pulse to ensure the health of the system. The windowing corelates to the physical distance between the Cal/BIT and the receive-antennas. After the initial power on of the system this is the first step that is required to be completed to set the monitor limits of the system. The following procedure will require that the system is installed to the siting requirements, and the system survey is completed, and all relevant information is entered to the system software.

Framing the Cal/BIT TOA Window

1. Press “Archive…” button on MI.

NOTE

Do not change the default file name.

2. Press “Run” button in Archive Dialog.
3. Press “Close” when process is complete to return to MI main window.
4. In SYSTEM SETUP>SETTINGS> DATA RECORDING, ensure “Record BIT RAM files” is checked (See Figure 6-1).
5. Select OK.
6. Close system setup window.

31. Open ESA BIT file from file pulldown in Multibit.
32. Examine TOA and TOA jitter in MultiBIT.
33. Enable Cal/BIT limits display and compare measurement data to monitor limits.
34. Determine F1 setting that resulted in lowest TOA jitter.

NOTE

- The setting used to compute the F1 Cal/BIT Position may not be the lowest point viewed from the individual bit file. The setting used will be lowest point at which **both** the ASA and ESA jitter is lowest. Example: if ASA jitter is lowest at .05 uS and ESA is lowest at .15 uS but the lowest point that both share is .20 uS, you would use .20 uS.
- To compute new F1 Position add the T_L to the determined setting, E.G…5.0uS + .20 uS = 5.2 uS.

35. Determine F1 setting that resulted in lowest TOA jitter for both ESA and ASA.

NOTE

TOA jitter should be between 2 to 4 ns. Record this best F1 setting in section 5 System Configuration Check Sheet.

36. Set F1 Cal/BIT Position to optimal setting (as determined in previous step) on MI, in "Settings."

END OF TASK